Exploring a Novel Pearson-like Scaled-Bregman Divergence for Multivariate Time Series Change-Point Detection: Mathematical Analysis and Application

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Motivation

The primary objective of change-point detection (CPD) is to identify specific time points where the underlying system undergoes transitions between different states, each characterized by its distinct data distribution. CPD has a number of applications across various real-world domains, including network analysis, motion detection, abnormal signal processing, identification of differentially expressed genes, and inference of time-varying regulatory networks.

Existing Approaches

- Parametric approaches: Gaussian mixture models, cumulative sum (CUSUM) methods, etc.
- Non-parametric approaches
  - Kernel-based Methods: kernel maximum mean discrepancy (MMD), maximum Kernel Fisher discriminant ratio-based method
  - Density-ratio Estimation Methods: Ratio matching, Kullback-Leibler(KL) importabce estimation procedure(KLIEP)

Methods

The goal of change-point detection is to learn the underlying probability distributions corresponding to the time series data both before and after the change-point. To avoid the pre-knowledge of data distribution, we applied density ratio estimation methods which focus on the direct estimation of the density ratio $\frac{p(y)}{q(y)}$, and prepared our Novel Pearson-like scaled-Bregman divergence (PLsBD) method.

- **Pearson-like scaled-Bregman divergence (PLsBD)** Given two measures P and Q on $\mathcal{Y}$ with densities p and q respectively, the separable Bregman divergences is defined as

$$D_{BR}(p\|q) = \int_{\mathcal{Y}} f(p(y)) - f(q(y)) - f'(q(y))(p(y) - q(y))dy,$$

where $f : \mathbb{R}^+ \to \mathbb{R}$ is a convex function, f' is its right derivative. Introducing a third measure M with density m leads to the definition of a scaled Bregman divergence, which encompasses a broad family of divergences. This divergence is expressed as:

$$D_{sBR}(p\|q;m) = \int_{\mathcal{Y}} f\left(\frac{p(y)}{m(y)}\right) - f\left(\frac{q(y)}{m(y)}\right) \quad (1)$$

$$- f'\left(\frac{q(y)}{m(y)}\right)\left(\frac{p(y)}{m(y)} - \frac{q(y)}{m(y)}\right) dM, \quad (2)$$

where $dM = m(y)dy$. Different choices of the measure M and function f yield various well-known divergence functions. In work we take $m(y) = q_\alpha(y)$

Theorem

Given the measure $m_\alpha(y) = \alpha p(y) + (1-\alpha)q(y)$, the Pearson-like scaled Bregman divergence can be expressed as $D_{PL}(p\|q;m_\alpha) = \frac{1}{2}E_p(r_\alpha(Y)) - \frac{2-\alpha}{2(1-\alpha)}E_q(r_\alpha(Y)) + \frac{1}{2(1-\alpha)}$, where $r_\alpha(Y) = \frac{p(Y)}{m_\alpha(Y)}$, and E_p and E_q denote the calculation of means with respect to the densities p and q, respectively.

- **Kernel-based Density Ratio Estimation**:The density ratio $r_\alpha(Y) = \frac{p(Y)}{m_\alpha(Y)}$ can be estimated using a Gaussian kernel regression method, which is modeled by

$$r_\alpha(Y;) = \sum_{l=1}^{n} \theta_l K(Y, Y_l), \quad (3)$$

where $= (\theta_1, \ldots, \theta_n)^\top$ are parameters learned from the data, and $K(Y, Y_l)$ is a Gaussian kernel expressed as:

$$K(Y, Y_l) = \exp\left(-\frac{\|Y - Y_l\|^2}{2\sigma^2}\right),$$

with a parameter σ (> 0) which can be estimated using cross-validation method.

Procedure Outline

Algorithm 1 Pearson-like scaled Bregman divergence-based change-points detection

Input: Time series data; α: mixture value in measure m_α; n: number of kernel functions; n_f: number of folds to use in cross validation; η: percentile threshold (0, 1)...

Step 1: Construct sliding windows
Divide time series data into consecutive segments Y^p, Y^q with length n_p and n_q.

Step 2: Identify optimal values of σ and λ
For each σ, λ value, apply n_f-fold cross validation to
Calculate the kernel $K(;\sigma)$,
Calculate $\widehat{\boldsymbol{H}}$ and $\widehat{\boldsymbol{h}}_p$ in Eq 2.11 and Eq 2.12,
Minimize the loss functions $J(\boldsymbol{\theta})$ in Eq 2.10 to identify optimal σ and λ.

Step 3: Estimation of density ratios
Calculate $\widehat{\boldsymbol{\theta}}$ and kernel $K(;\sigma)$ using optimal σ and λ,
Calculate $\widehat{r_\alpha}(Y;\boldsymbol{\theta}) = \sum_{\ell=1}^{n} \widehat{\theta}_\ell K(Y, Y_\ell)$.

Step 4: Estimation of Pearson-like scaled Bregman divergence (PLsBD)
Estimate $\hat{D}_{PL}(p\|q;m_\alpha) \approx \frac{1}{2n_p}\sum_{i=1}^{n_p}(\hat{r}_\alpha(Y_i^p)) - \frac{2-\sigma}{2(1-\alpha)n_q}\sum_{i=1}^{n_q}(\hat{r}_\alpha(Y_i^q)) + \frac{1}{2(1-\alpha)}$.
Move the sliding windows forward to repeat the above steps until the end of data.

Step 5: Identify change-points using PLsBD score
Sort the PLsBD score $\hat{D}_{PL}(p\|q;m_\alpha)$ and select the change-point candidates according to the percentile η

Output: Multiple change-points.

Application and Result

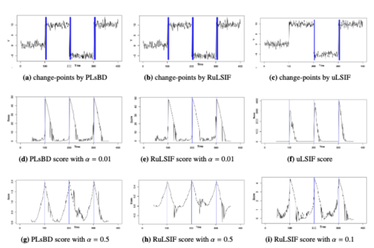
Datasets

- Synthetic Dataset
  - **Dataset 1: Time series Gaussian signal.** The synthetic time-series Gaussian signal consists of four segments of data with an equal length of 100, and the data are generated from four different normal distributions: $\mathcal{N}(0,1)$, $\mathcal{N}(10,1)$, $\mathcal{N}(-5,1)$, and $\mathcal{N}(10,1)$, respectively. Thus, the changes of means occur at time 101, 201, and 301, which are the predefined change-points.
  - **Dataset 2: auto-regressive model.** The 1-dimensional auto-regressive model AR(2), is used to generate 1,000 samples ($t = 1, \ldots, 1,000$):

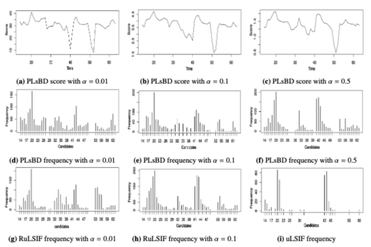
$$Y_t = 0.6Y_{t-1} - 0.5Y_{t-2} + \epsilon_t,$$

where $\epsilon_t \sim \mathcal{N}(0,1)$. The initial values are set as $Y_1 = Y_2 = 0$. A change-point is inserted at every 100 time steps by increasing the noise mean μ by 2, that is, $\mu_t = \mu_{t-100} + 2$, where $t = 100N + 1$, where $N = 1, \ldots, 9$, that is, there are 9 change-points.
- Real-world high-dimensional microarray data
  - Arbeitman *et al.*'s study measured the gene expression levels of 4028 genes of wild-type Drosophila across 67 successive time points throughout its life cycle: 31, 10, 18, and 8-time points for the embryo, larva, pupa, and adult stages, respectively. So, the four major stages of morphogenesis correspond to the change-points around the time points 32, 42, and 60.
  - Microarray data comprising 4028 genes of the Drosophila life cycle: we repetitively and randomly select 40 genes out of the 4028 genes for 1000 iterations, maintaining fixed parameter values as in the Drosophila dataset.

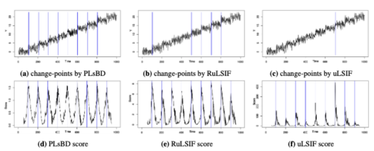
Result



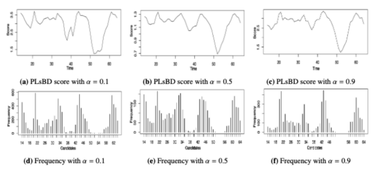
(a) Gaussian signal's change-point detection using the Pearson-like scaled Bregman divergence (PLsBD), RuLSIF, and uLSIF methods.



(b) Change-points inferred from time series microarray data of eleven genes involved in Drosophila's muscle development.



(c) Change-point detection within time series data generated by the Auto-Regressive model AR(2).



(d) Change-points inferred from time series microarray data of 4028 genes in Drosophila's life cycle.